Canvas3D-v2: Scene-Graph-Grounded Viewpoint-Controllable Painting Animation with Spatial-Relation-Driven Video Synthesis

Anonymous Authors
AIforEducation Project
aiforeducation@example.com

Abstract

Animating static artworks—oil paintings, ink-wash scrolls, and frescoes—into explorable, narratively coherent videos is a long-standing goal at the intersection of computer vision, generative modeling, and digital cultural heritage. Existing approaches remain bifurcated: image-to-video diffusion models produce visually fluent motion but collapse three-dimensional structure under viewpoint change, while feed-forward 3D reconstruction pipelines capture geometry at the cost of obliterating painterly texture, impasto, and brushstroke identity. The root cause is a fundamental *objective tension*: artistic style fidelity and 3D scene structure are governed by incompatible optimization targets, and no single end-to-end model satisfies both, especially when the task demands object-level manipulation. We propose **Canvas3D-v2**, a framework that resolves this tension through a *modular decomposition of concerns*. Rather than training a monolithic network, Canvas3D-v2 factors the painting-to-video problem into seven independently optimized, mutually reinforcing sub-problems: *(i) perceptual grouping* of a painting into semantically meaningful foreground objects and a preserved background, jointly with monocular depth-derived layer ordering; *(ii) per-object geometric lifting* that converts each 2D region into a metric-scaled 3D representation and assembles a reusable, retrievable object-level model library; *(iii) line-art scene composition* in which hand-drawn sketches serve as *spatial constraints* rather than stylistic hints, solved by a contour-based rigid registration that places each 3D object at its sketched location and a depth-conditioned inpainting module that restores disoccluded background texture in a geometry-aware manner; *(iv) 3D scene graph construction* that organizes the lifted objects and their pairwise spatial relations—topological, depth-ordering, and physical-support—into an explicit, editable, serializable graph representation, providing a structured intermediate that subsequent viewpoint and video stages can reason over; *(v) viewpoint-controllable novel-view rendering* that accepts a user camera trajectory specification (manual, orbit, dolly, crane, or auto-orbit) and renders multi-view keyframes with per-node visibility masks, enabling arbitrary viewpoint exploration rather than a fixed arc; and *(vi) spatial-relation-driven video synthesis* that injects a third control signal—a graph-neural-network-encoded relation embedding pooled per-frame over the visible node set—into the video diffusion backbone alongside depth and edge, anchoring temporal motion to explicit 3D *spatial relations* rather than learned heuristics, with a relation-consistency verification loop that re-prompts when generated frames violate the edited scene graph. The key insight is that *separating perceptual, geometric, compositional, relational, viewpoint, and temporal concerns* turns an ill-posed, conflicting joint problem into seven well-posed sub-problems, each addressable by an off-the-shelf specialist, while their interaction yields capabilities—arbitrary object repositioning, viewpoint-controllable exploration, and spatial-relation-anchored animation—that no single-modality pipeline offers. On a curated benchmark of classical paintings, Canvas3D-v2 achieves state-of-the-art style fidelity, perceptual quality, temporal coherence, novel-view consistency, and spatial-relation accuracy, with user studies confirming strong preference for artistic authenticity, motion naturalness, compositional-intent alignment, and viewpoint-exploration quality.

1 Introduction

Classical paintings—whether oil on canvas, ink on silk, or pigment on fresco—are cultural artifacts that encode centuries of human visual imagination Lowenthal [2015]. From van Gogh's swirling *Starry Night* to Zhang Zeduan's panoramic *Along the River During the Qingming*

Festival, these works imply motion, narrative, and spatial depth that their static medium can only suggest. Animating such paintings as explorable videos has therefore emerged as a compelling research goal at the intersection of computer vision, generative modeling, and digital cultural heritage Zhang *et al.* [2024]; Dong [2024].

Recent years have witnessed remarkable progress on two fronts that, however, fail to meet on common ground. On one hand, image-to-video (I2V) diffusion models Blattmann *et al.* [2023]; Wan Team [2025]; Lightricks Team [2025]; Yang *et al.* [2025], combined with controllable conditioning mechanisms Zhang *et al.* [2023a]; Wang *et al.* [2024c]; Peng *et al.* [2024], can animate a single photograph into a short clip with plausible per-frame dynamics. Yet when applied directly to oil paintings or ink-wash artworks, they suffer from two fundamental shortcomings. First, they operate entirely in 2D pixel space: the model has no geometric understanding of object depth ordering, self-occlusion, or relative spatial layout, and consequently cannot synthesize novel viewpoints or maintain consistent 3D structure across frames. Second, long-range temporal coherence degrades rapidly beyond 2–4 seconds, as the latent-space interpolation lacks explicit geometric constraints to anchor the motion trajectory.

On the other hand, 3D reconstruction and rendering methods Mildenhall *et al.* [2020]; Kerbl *et al.* [2023]; Huang *et al.* [2024] can reconstruct explorable 3D scenes from sparse views, and feed-forward monocular variants Wang *et al.* [2024e]; Tang *et al.* [2024]; Charatan *et al.* [2024] can even lift a single image to a 3D representation. However, these reconstruction pipelines are optimized for photorealistic fidelity; when applied to paintings, the rendering process replaces expressive brushstrokes with geometric artifacts, flattens the impasto texture, and bleeds semantic color boundaries under novel-view projection.

The core tension is this: *3D scene structure and artistic style fidelity* are governed by incompatible optimization objectives, and no single model family—neither pure diffusion nor pure 3D reconstruction—satisfactorily achieves both. This tension becomes especially acute when the animation task requires **object-level manipulation**: moving a painted figure from one compositional position to another while preserving its painterly identity and filling the resulting gaps with plausible background content.

Our approach: modular separation of concerns. Rather than training a single monolithic network to jointly optimize style and structure—an ill-posed objective whose gradients pull in opposite directions—we propose to *factor* the painting-to-video problem into seven interacting yet independently optimized sub-problems, each addressed by a specialist and communicating with the next through a small, interpretable handshake representation (Figure 1):

- **PaintParse** handles *perceptual grouping*: decomposing the input painting into foreground objects and a background layer using an art-domain-adapted segmentation backbone, augmented with monocular depth estimation for metric-scale layer ordering.
- **MonoSplat** handles *geometric structure*: lifting each segmented foreground region into a high-fidelity 3D representation via feed-forward single-image 3D prediction, assembling an object-level model library with depth-derived metric rescaling.
- **SketchComp** handles *compositional intent*: accepting hand-drawn line-art sketches specifying target object positions, solving a contour-based rigid registration to place each 3D object onto its sketch, and filling disocclusion gaps through a depth-conditioned inpainting module that preserves background artistic texture.
- **SceneGraph3D** handles *spatial-relational structure*: organizing the lifted 3D objects and the background into an explicit, editable 3D scene graph whose edges encode pairwise topological, depth-ordering, and physical-support relations, providing a structured intermediate that the viewpoint and video stages can reason over.
- **ViewCtrl** handles *viewpoint control*: accepting a user camera trajectory specification (manual, orbit, dolly, crane, or auto-orbit around the most salient node) and rendering a multi-view keyframe stream with per-node visibility masks, enabling arbitrary viewpoint exploration rather than a fixed arc.
- **SpatialFlow** handles *spatial-relation-driven video synthesis*: injecting a graph-neural-network-encoded relation embedding—pooled per-frame over the visible node set—into the video diffusion backbone alongside depth and edge, anchoring temporal motion to explicit 3D spatial relations, with a relation-consistency verification loop that re-prompts when generated frames violate the edited scene graph.
- **ArtiFlow** handles *final temporal rendering*: compositing the viewpoint keyframes and the three control signals (depth + edge + relation) into the video diffusion backbone with first-frame latent anchoring for pixel-precise object motion.

This decomposition enables a clean separation of concerns, and the handshakes carry exactly the information each downstream stage needs and no more, preventing error compounding. The result is a system that supports *arbitrary object repositioning*, *viewpoint-controllable exploration*, *spatial-relation-anchored animation*, and *style-anchored rendering*—capabilities that no prior single-modality pipeline offers.

Contributions. Our contributions are:

1. **A modular factorization** of the painting-to-video problem into seven independently optimized specialists—PaintParse (perceptual grouping), MonoSplat (per-object 3D lifting), Sketch-

Comp (line-art composition with depth-aware inpainting), SceneGraph3D (3D scene graph construction), ViewCtrl (viewpoint-controllable rendering), SpatialFlow (relation-driven video synthesis), and ArtiFlow (final video composition)—communicating through interpretable handshake representations. To our knowledge this is the first end-to-end pipeline for 3D-aware, viewpoint-controllable, and spatial-relation-anchored painting animation.

2. **An object-level 3D lifting pipeline** (MonoSplat) that combines art-domain-adapted instance segmentation with feed-forward monocular 3D prediction and depth-derived metric rescaling, producing a reusable, retrievable library of 3D foreground objects from a single painting.

3. **A sketch-conditioned 3D scene composition algorithm** (SketchComp) with a contour-based rigid registration that jointly optimizes region IoU and boundary Chamfer distance, and a depth-aware color inpainting module that fills disocclusion gaps while preserving the background's artistic texture.

4. **A 3D scene graph representation** (SceneGraph3D) that encodes pairwise spatial relations—topological (inside, adjacent, occludes), depth-ordering (in-front-of, behind), and physical-support (supports, leans-on)—as an editable, serializable graph, providing a structured intermediate for downstream viewpoint and video reasoning.

5. **A viewpoint-controllable novel-view rendering module** (ViewCtrl) that supports user-specified camera trajectories (manual, orbit, dolly, crane, auto-orbit) and produces per-node visibility masks, enabling arbitrary viewpoint exploration while preserving brushstroke identity.

6. **A spatial-relation-driven video synthesis workflow** (SpatialFlow) that injects a GNN-encoded relation embedding as a third control signal alongside depth and edge into the video diffusion backbone, with per-frame visibility-aware pooling and a relation-consistency verification loop that re-prompts when generated frames violate the edited scene graph.

7. **Comprehensive evaluation** on a curated benchmark of 50 classical oil paintings and ink-wash works, achieving state-of-the-art style fidelity, perceptual quality, temporal coherence, novel-view consistency, and spatial-relation accuracy, with user studies confirming strong preference for artistic authenticity, motion naturalness, compositional-intent alignment, and viewpoint-exploration quality.

2 Related Work

We organize prior work along the four axes that motivate our modular decomposition, tracing the developmental lineage of each thread and positioning Canvas3D at their intersection.

2.1 From Pixel-Level Stylization to Painting Animation

The earliest approaches to painting animation treat the input as a *pixel grid to be stylized over time*. Static image style transfer, pioneered by feature-correlation methods Gatys *et al.* [2016] and extended by fast feed-forward networks Johnson *et al.* [2016]; Huang and Belongie [2017] and transformer-based stylizers Deng *et al.* [2022], established that artistic appearance can be re-injected into arbitrary content. Video style transfer then addressed temporal coherence through optical-flow regularization Ruder *et al.* [2016]; Chen *et al.* [2017]; Wang *et al.* [2020], multi-channel correlation Deng *et al.* [2021], and content deformation fields Ouyang *et al.* [2024]. A second lineage pursues *stroke-level synthesis*—reproducing the physical painting process as a time-lapse Zhao *et al.* [2020]; Zou *et al.* [2021]; Liu *et al.* [2021]—but such methods reconstruct *how a painting was made* rather than *how its subjects could move*. A third, narrower thread drives *portrait motion* on painted faces Guo *et al.* [2024]; Tan *et al.* [2024] but cannot animate full-body figures or background elements. The diffusion era introduced inversion-based stylization Zhang *et al.* [2023b] and video-consistent editing Ceylan *et al.* [2023]; Chai *et al.* [2023]; Geyer *et al.* [2024]; Liang *et al.* [2024], which preserve appearance under text-driven edits but provide no geometric handle on object pose. The recent InkSpirit Zhang *et al.* [2025] and StyleMaster Ye *et al.* [2025] push appearance fidelity further yet remain confined to 2D image translation. Across this lineage, the persistent gap is the absence of *object-level, 3D-grounded motion control*: appearance is preserved, but spatial structure is not.

2.2 3D Scene Representations and Single-Image Lifting

Three-dimensional representations evolved from implicit radiance fields Mildenhall *et al.* [2020] to explicit primitives. 3D Gaussian Splatting Kerbl *et al.* [2023] represents a scene as a set of anisotropic Gaussians with differentiable rasterization, enabling real-time photorealistic rendering; 2D Gaussian Splatting Huang *et al.* [2024] collapses the third scale dimension into a surface normal, improving geometric fidelity for surface-like content. Crucially, the field has recently shifted from *per-scene optimization* to *feed-forward single-image lifting*: generative-prior methods Tang *et al.* [2024] predict 3D assets from one image, while state-of-the-art feed-forward predictors Wang *et al.* [2024e]; Charatan *et al.* [2024]; Jiang *et al.* [2025] regress Gaussian parameters directly, with two-view and unconstrained-view variants broadening the input regime. These advances make it feasible to lift individual painting regions into 3D, but none addresses the downstream challenge of *composing multiple such objects into a coherent, artistically controllable scene*—exactly the gap our framework targets.

2.3 Sketch and Line-Art as a Composition Interface

Hand-drawn sketches have a long history as an intuitive interface for shape prototyping Olsen *et al.* [2009]. Recent work demonstrates that contour-level spatial reasoning is tractable with modern vision models: ShadowDraw Luo *et al.* [2026] generates compositional shadow art from object inputs, LineArt Wang *et al.* [2024d] performs training-free appearance transfer onto design drawings via diffusion, and region-aware simplification Nguyen *et al.* [2024] and volumetric animation toolkits Fox-Gieg [2024] illustrate the expressive potential of hand-drawn control. In diffusion-based generation, sketches typically serve as a *stylistic* or *boundary* condition (e.g., ControlNet-style edge injection Zhang *et al.* [2023a]). Canvas3D departs from this convention by treating line-art as a *spatial constraint* that resolves the which-object-where problem in 3D scene assembly: a sketch is not merely an appearance hint but a target silhouette that a 3D object is geometrically registered to.

2.4 Controllable Image-to-Video Generation

The leap from text-to-image to text-to-video was powered by diffusion transformers Peebles and Xie [2023] and video diffusion models Ho *et al.* [2022a,b]. A family of scalable backbones Yang *et al.* [2025]; Wan Team [2025]; Lightricks Team [2025]; Blattmann *et al.* [2023] now supports high-fidelity image-to-video synthesis. Controllability has been pursued through auxiliary conditioning: ControlNet Zhang *et al.* [2023a] injects edge, depth, and pose maps into the diffusion U-Net, with video extensions Wang *et al.* [2024c] and efficiency-oriented variants Peng *et al.* [2024] broadening its applicability. Cinematic shot design Xing *et al.* [2025] and clipart animation Wu *et al.* [2025] demonstrate compelling 3D-aware control without explicit 3D supervision. Despite these advances, existing I2V control methods are confined to *per-frame conditioning signals*: they cannot reason about 3D object identity, depth ordering, or occlusion across frames, and consequently drift under viewpoint change. Canvas3D addresses this by grounding all control signals in an explicit, object-level 3D scene that is composited, rendered, and then fed into the video diffusion backbone.

2.5 Painting-Specific Animation Pipelines

A small but growing body of work targets paintings directly. Domain-adapted detectors Chen *et al.* [2025] and art-segmentation datasets Saleh and Elgammal [2016] establish that natural-image segmentation backbones Jocher *et al.* [2023]; Wang *et al.* [2024b,a]; Tian *et al.* [2025] and foundation segmentation models Kirillov *et al.* [2023]; Ravi *et al.* [2024] require art-specific fine-tuning due to brushstroke texture and intentional boundary blending. Chinese ink-wash style transfer has been explored via instance-aware coherence Liang *et al.* [2021] and staged diffusion guidance Liu *et al.* [2026]. Most relevant to our setting, "Printing the Underdetermined" Ming *et al.* [2026] translates paintings to 3D representations via video generative models but operates on the entire image jointly rather than at the object level, while OilPaintingRevive Anonymous [2026] proposes a multi-stage pipeline combining 2DGS with video refinement, but its skeleton-binding network requires per-scene optimization that limits compositional flexibility. Canvas3D advances beyond these works along four axes—*object-level* rather than image-level manipulation, *line-art* as a compositional control interface, *depth-aware inpainting* for disocclusion, and *multi-modal geometric conditioning* for video synthesis—yielding stronger 3D consistency and explicit compositional control.

3 Method

Design philosophy. A single painting-animation system must simultaneously satisfy four *conflicting* objectives: perceptual grouping of artistically rendered content, geometric reconstruction of objects that have no real 3D extent, user-specifiable composition, and temporally coherent stylized rendering. Jointly optimizing a single network for all four is ill-posed because the gradients of style fidelity and 3D structure pull in opposite directions. Our central design principle is therefore *modular separation of concerns with explicit handshakes*: each objective is addressed by an independently optimized specialist, and the four specialists communicate only through well-defined, interpretable intermediate representations—masks with depth-ordered layers, a metric-scaled object-level 3D library, a composited 3D scene with disocclusion-filled background, and finally a multi-view keyframe stream with paired geometry and edge cues. This decomposition converts a conflicting joint problem into four well-posed sub-problems, while the handshakes carry exactly the information each downstream stage needs and no more, preventing error compounding.

Problem statement. Given a painting scan $P \in \mathbb{R}^{H\times W\times 3}$ and a set of hand-drawn line-art sketches $L = \{\ell_1, \ldots, \ell_K\}$ specifying target object positions, Canvas3D-v2 produces an animated video V that (i) preserves the artistic style of P, (ii) supports user-specified viewpoint exploration of a 3D-composited scene, (iii) places each foreground object at its sketched location with geometrically grounded, temporally coherent motion, and (iv) respects user-edited spatial relations across frames. The framework comprises seven modules (Figure 1): **PaintParse** (perceptual grouping and layer decomposition), **MonoSplat** (per-object 3D lifting and library assembly), **SketchComp** (line-art-guided scene composition with depth-aware inpainting), **SceneGraph3D** (explicit 3D scene graph construction, §3.6), **ViewCtrl** (viewpoint-controllable novel-view rendering, §3.7), **SpatialFlow** (spatial-relation-driven video synthesis, §3.8), and **ArtiFlow** (final video composition with first-frame latent anchoring). The first four modules form the base pipeline shown in Figure 1; the three

v2 modules build on their outputs.

Notation. $M_i \in \{0,1\}^{H\times W}$ denotes a binary mask for the i-th foreground object, M_{bg} the residual background mask, D_{mono} the monocular depth map, $\mathcal{G}_i$ the 3D representation of object i, $\mathcal{S}$ the composited scene, Ω_{gap} the disocclusion region, and $\{I_t, D_t, E_t\}$ the multi-view RGB, depth, and edge keyframes.

3.1 PaintParse: Art-Domain Perceptual Grouping and Layer Ordering

Design rationale. The downstream 3D and composition stages require two distinct outputs that no single head of an off-the-shelf detector provides: *(i)* foreground masks whose boundaries respect artistic intent rather than photographic edges, and *(ii)* a depth-consistent layer ordering that disambiguates which object lies in front of which. PaintParse therefore couples a domain-adapted instance segmentation backbone with a parallel monocular depth estimator and fuses their outputs through a percentile-based layer assignment. The key design choice is *decoupling semantic grouping from depth ordering*: the segmentation network is responsible only for "what is an object," while the depth estimator is responsible only for "how far away it is," and the layer-assignment logic simply reads both off. This avoids asking either network to solve a problem it was not trained for.

Instance segmentation with art-domain adaptation. We adopt a one-stage instance segmentation backbone pre-trained on natural-image data. Direct transfer to paintings yields poor mask quality because brushstroke textures produce spurious edge responses and depth-ambiguous compositions confound object-background separation. We mitigate this through a two-stage adaptation protocol: a *low-rank adaptation* of the detection backbone to inject art-domain statistics without catastrophic forgetting, followed by *full fine-tuning* of the mask-prediction head to align boundary prediction with painterly edges. Given input P, the adapted model outputs K binary masks $\{M_i\}_{i=1}^{K}$. The residual background mask is computed by pixel-wise subtraction:

$$M_{\text{bg}}(p) \;=\; 1 \;-\; \max_{i\in\{1,\dots,K\}} M_i(p), \quad \forall p \in [1,H]\times[1,W]. \tag{1}$$

Monocular depth and layer assignment. In parallel, a monocular depth estimator produces a relative depth map D_{mono}. For each detected object i, we summarize its depth by the 10th and 90th percentiles of $\{D_{\text{mono}}(p) \mid p \in \Omega_i\}$ where $\Omega_i = \{p \mid M_i(p){=}1\}$, yielding a robust depth range $[d_i^{\text{near}}, d_i^{\text{far}}]$ that is insensitive to outlier pixels at mask boundaries. The layer index of each object is then determined by the depth percentile of its centroid, providing the depth-ordered layering that downstream composition requires. This percentile-based assignment is preferable to a per-pixel depth sort because it treats each object as an atomic unit, preserving its identity under subsequent rigid transformation.

3.2 MonoSplat: Per-Object 3D Lifting and Library Assembly

Design rationale. A whole-scene 3D reconstruction would entangle object geometry with background texture, making independent repositioning impossible. We therefore lift each foreground object *separately* into a 3D representation and assemble a *reusable library*, indexed by semantic label and original painting coordinates. Two design choices follow from this principle. First, we use a *feed-forward single-image 3D predictor* rather than per-scene optimization, because per-object optimization would be prohibitively expensive and would overfit to a single viewpoint. Second, because feed-forward predictors operate in a normalized canonical space without metric scale, we *recover metric dimensions from the PaintParse depth cue*, so that objects from different paintings can later be composed in a common metric frame.

Feed-forward lifting. For each foreground object i, we crop the painting to its bounding box, $P_i = P \odot M_i$, and feed the crop into a feed-forward single-image 3D predictor that regresses a set of N_i 3D Gaussians directly from one image. Each Gaussian is parameterized by a center $\boldsymbol{\mu}_j \in \mathbb{R}^3$, axis-aligned scales $\mathbf{s}_j \in \mathbb{R}^3_+$, rotation quaternion $\mathbf{q}_j \in \mathbb{R}^4$, opacity $\alpha_j \in [0,1]$, and view-dependent color coefficients. The predictor's generative prior, trained on diverse 3D assets, generalizes to the missing side and back of objects that are not visible in the painting—a critical capability for downstream rotation during composition.

Metric rescaling. The predictor outputs in a normalized canonical space. We recover the metric scale using the depth cue from PaintParse:

$$\lambda_i \;=\; \frac{d_i \cdot h_i^{\text{px}}}{f}, \tag{2}$$

where d_i is the median depth of object i, h_i^{px} is its pixel height, and f is the focal length in pixels. Each Gaussian center and scale is then rescaled by λ_i, producing a metric 3D object. The resulting library $\{\mathcal{G}_1, \dots, \mathcal{G}_K\}$ is keyed by semantic label and original painting coordinates, enabling selective retrieval and cross-painting composition in future work.

3.3 SketchComp: Line-Art-Guided 3D Scene Composition

Design rationale. The user's compositional intent—which object goes where—is most naturally expressed as a sketch, but existing diffusion-based sketch control treats sketches as appearance or boundary cues rather than spatial constraints. SketchComp inverts this convention: a sketch *is* the target, and a 3D object is *geometrically registered* to it. Two sub-problems must then be solved: *(i)* finding the rigid transformation that aligns each 3D object's rendered silhouette with its sketched contour, and *(ii)* filling the disocclusion gaps that open up when an object moves away from its original position,

while preserving the background's artistic texture. We solve the first with a silhouette-overlap objective that combines region IoU with boundary Chamfer distance, and the second with a depth-conditioned inpainting module whose geometric conditioning prevents the texture-smear artifact that naive inpainting produces on paintings.

Contour-based rigid registration. Each sketch ℓ_i is converted to a binary edge map via Canny detection, and its external contour Γ_i is extracted. For each 3D object $\mathcal{G}_i$, we search for the rigid transformation—rotation θ, 2D translation $\mathbf{t}$, and uniform scale s—that best aligns the rendered silhouette with Γ_i:

$$(\theta_i^*, \mathbf{t}_i^*, s_i^*) = \arg\min_{\theta,\mathbf{t},s} \mathcal{L}_{\text{cont}}(\theta, \mathbf{t}, s), \tag{3}$$

$$\mathcal{L}_{\text{cont}} = \underbrace{1 - \text{IoU}\big(\mathcal{R}_i, \Gamma_i\big)}_{\text{region}} + \lambda_c \underbrace{\text{Chamfer}\big(\partial\mathcal{R}_i, \Gamma_i\big)}_{\text{boundary}},$$

where $\mathcal{R}_i = \mathcal{R}(\mathcal{G}_i; \theta, \mathbf{t}, s)$ is the silhouette rendered from the canonical camera and λ_c balances the two terms. The IoU term rewards global region agreement, while the Chamfer term penalizes boundary misalignment—their combination avoids the degenerate solutions that either term alone admits (a small silhouette inside the contour maximizes IoU locally; boundary-only matching can produce topologically wrong regions). Optimization proceeds via gradient descent with iterative re-rendering, so that each step's gradient is computed against the current geometry rather than a stale approximation.

Scene assembly and disocclusion inpainting. Each registered object is placed at its PaintParse-derived depth d_i and objects are rendered in depth-sorted order against a background layer reconstructed from the RGB-D map. Object displacement creates *disocclusion gaps*—background pixels newly visible after objects move—which we detect as

$$\Omega_{\text{gap}} \;=\; \big\{\, p \in M_{\text{bg}}^{\text{orig}} \;\big|\; \text{no 3D surface projects to } p \,\big\}. \tag{4}$$

Filling Ω_{gap} while preserving artistic texture coherence is critical. We employ a depth-conditioned inpainting diffusion pipeline: a masked composite (with Ω_{gap} zeroed) is fed into a latent inpainting diffusion model with a depth-conditioned control branch conditioned on the composited depth map. The depth conditioning is the key design choice: it constrains the inpainted content to follow the background's geometric structure, preventing the texture-smear artifact that arises when an unconstrained inpainter fills a painting gap with semantically plausible but geometrically inconsistent content. A dilation radius expands the surrounding context to supply texture reference, ensuring the filled region blends with adjacent brushwork. The resulting scene $\mathcal{S}^*$ and its depth map $\mathcal{D}^*$ are forwarded to ArtiFlow.

3.4 ArtiFlow: Geometry-Grounded Video Synthesis

Design rationale. Pure image-to-video diffusion applied to a painting produces plausible per-frame motion but suffers two failure modes: geometric drift (objects wander off their intended trajectories) and boundary melting (brushstroke edges blur across frames). ArtiFlow addresses both by *grounding the diffusion in explicit 3D geometry*: the composited scene is rendered into a multi-view keyframe stream that provides three control signals—RGB appearance, depth geometry, and edge boundary—which are injected into a video diffusion backbone as dual control branches. The depth branch anchors each frame to the 3D structure of $\mathcal{S}^*$, suppressing geometric drift; the edge branch preserves brushstroke-level boundary detail, suppressing boundary melting. The combined conditioning converts the diffusion backbone from a free-form motion generator into a *geometry-anchored refiner* whose temporal freedom is restricted to the residual motion that the 3D rendering does not already specify.

Multi-view keyframe rendering. Given $\mathcal{S}^*$, we define a smooth camera trajectory through T poses $\{\mathbf{T}_t\}_{t=1}^{T}$, generated by cubic B-spline interpolation between a canonical front-facing start pose and a user-specified or auto-generated end pose. At each step t, differentiable rasterization produces an RGB frame $I_t = \mathcal{R}_{\text{rgb}}(\mathcal{S}^*, \mathbf{T}_t)$, a depth map $D_t = \mathcal{R}_{\text{depth}}(\mathcal{S}^*, \mathbf{T}_t)$, and an edge map $E_t = \text{Edge}(I_t)$. The keyframe count is set to $T = 8N + 1$ to align with the temporal chunk constraints of modern video diffusion backbones.

Dual-conditioned video diffusion. Two parallel control branches inject geometric and boundary cues into the video diffusion backbone. The depth branch, conditioned on $\{D_t\}$ at strength s_d, anchors each frame to the 3D structure of $\mathcal{S}^*$, preventing the geometric drift that plagues pure I2V models. The edge branch, conditioned on $\{E_t\}$ at strength s_e, preserves brushstroke-level boundary detail, suppressing the boundary-melting artifact common when diffusion is applied to paintings. The combined conditioning signal per frame is

$$\mathbf{c}_t \;=\; s_d\, \mathbf{c}_t^{\text{depth}} \;+\; s_e\, \mathbf{c}_t^{\text{edge}}, \tag{5}$$

injected at the corresponding residual connections of the backbone. Temporal coherence is further reinforced by *first-frame latent anchoring*: subsequent frames' initial noise is set to the first-frame latent plus a position-encoded Gaussian perturbation $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \sigma_t^2 \mathbf{I})$ with σ_t growing linearly with $|t-1|$, so that motion freedom increases smoothly away from the anchored first frame.

Per-object motion verification. After generation, we verify motion fidelity by computing the IoU between rendered object masks and segmented masks in each generated frame. If the average IoU falls below a threshold τ_{mo}, the pipeline triggers a *re-composition* step (dashed arrow in Figure 1) that adjusts control strengths or regenerates the affected segment with stronger depth conditioning, closing the loop between composition and synthesis.

3.5 Pipeline Algorithm and Implementation

Algorithm 1 summarizes the full Canvas3D pipeline. The modular design enables *fault-tolerant resumption*: each stage saves its outputs as checkpoint files, so any specialist can be upgraded or replaced without re-running downstream stages—for example, swapping in a stronger 3D predictor or a different video backbone. Peak memory consumption across all four stages is kept within a single 24 GB budget through serial model loading with explicit memory release between stages, which is feasible precisely because the handshake representations between stages are small, serializable tensors rather than shared activations.

3.6 SceneGraph3D: Explicit 3D Scene Graph Construction

Design rationale. The original four modules produce a composited 3D scene $\mathcal{S}^*$ but leave its *relational structure* implicit: the only relation between two objects is their depth ordering, encoded as a sort key. This suffices for fixed-camera animation but is insufficient for two new capabilities we target—*viewpoint-controllable exploration* and *spatial-relation-anchored video synthesis*—both of which require reasoning about which object is where, in front of what, and supporting or occluding which other object. SceneGraph3D lifts the implicit depth-sorted scene into an *explicit, editable, serializable 3D scene graph* $\mathcal{G}_{\text{scene}} = (\mathcal{V}, \mathcal{E})$, whose nodes carry per-object geometric and semantic attributes and whose edges carry typed, quantified spatial relations. The graph is the single handshake representation consumed by both ViewCtrl (§3.7) and SpatialFlow (§3.8), and is the locus of user *spatial-relation editing*—the user modifies the graph, not the pixels, and downstream modules re-render and re-generate accordingly.

Nodes and attributes. Each foreground object i from PaintParse and a single background node b become graph nodes. Each node carries: (i) semantic attributes—category label, salience $s_i = 0.5\tilde{a}_i + 0.3c_i + 0.2(1 - \tilde{r}_i)$, combining normalized mask area $\tilde{a}_i$, detection confidence c_i, and normalized center offset $\tilde{r}_i$; (ii) geometric attributes—metric 3D center $\boldsymbol{\mu}_i$, axis-aligned 3D bounding box, volume, height, and depth range $[d_i^{\text{near}}, d_i^{\text{far}}]$ from the MonoSplat metric rescaling; and (iii) rendering handles—the 3DGS model path and original 2D mask. The scene extent $\boldsymbol{\xi} = (\xi_x, \xi_y, \xi_z)$ is the outer bounding box of all foreground nodes and defines the spatial scale against which relation thresholds are normalized, making the graph scale-invariant.

Typed spatial relations. For every ordered pair of foreground nodes (i, j), $i \neq j$, SceneGraph3D infers up to four typed relations:

- *2D projection* (LEFT-OF, RIGHT-OF, ABOVE, BELOW): comparing the x, y components of $\boldsymbol{\mu}_i$ and $\boldsymbol{\mu}_j$ against $\varepsilon_{xy} = 0.1\,\xi_x$, with confidence proportional to the normalized offset.
- *Depth ordering* (IN-FRONT-OF, BEHIND): if $d_i^{\text{far}} < d_j^{\text{near}}$ then i is IN-FRONT-OF j with confidence $(d_j^{\text{near}} - d_i^{\text{far}})/\xi_z$; otherwise the centroids decide with confidence 0.5.
- *Topological* (INSIDE, ADJACENT, OCCLUDES): computed from the 2D masks. OCCLUDES fires when mask overlap exceeds 0.05 and i is in front of j; INSIDE when IoU > 0.9 and coverage > 0.8; ADJACENT when masks are disjoint but within a 5-pixel dilation.
- *Physical support* (SUPPORTS, LEANS-ON): a heuristic that fires when i's bbox bottom is within $0.05\,\xi_y$ of j's bbox top and i is above j, capturing the painterly convention that one object rests on another.

Each relation edge stores a confidence in $[0, 1]$ and, where applicable, a quantified attribute—3D Euclidean distance, mask overlap ratio, or depth gap—so that downstream reasoning can be soft rather than categorical. Every foreground node is additionally connected to the background node by an IN-FRONT-OF edge with confidence 0.95. The graph is serialized to JSON, making it a checkpointable, editable artifact.

Scene graph editing. The graph exposes a small editing API (MOVE, ROTATE, SCALE, ADD/REMOVE/CHANGE-RELATION) that SpatialFlow consumes. This is the key affordance that makes *spatial-relation controllable* video possible: the user expresses "move the person to the left of the building and behind the tree" as a graph edit, not a text prompt, and the edit propagates deterministically through ViewCtrl and SpatialFlow rather than being interpreted stochastically by a language model.

3.7 ViewCtrl: Viewpoint-Controllable Novel-View Rendering

Design rationale. ArtiFlow in the original Canvas3D renders keyframes along a fixed cubic-B-spline arc with a small amplitude, which suffices for parallax-rich "3D-ish" animation but does not let the user *explore* the painting from a chosen viewpoint. ViewCtrl promotes the camera from a fixed trajectory to a *user-specified specification*, decoupling viewpoint control from video synthesis: the user declares where to look from, ViewCtrl renders the multi-view keyframe stream, and SpatialFlow/ArtiFlow animate between them. Two design choices follow. First, viewpoint is specified in *world coordinates against the scene graph*, not in pixel offsets, so the same trajectory spec portably applies to any painting whose SceneGraph3D has been built. Second, each rendered frame carries *per-node visibility masks* recording which foreground objects are in the frustum and unoccluded, because SpatialFlow's relation embedding is pooled over the visible node set and must know which nodes contribute.

Trajectory specification. A viewpoint trajectory is a list of camera poses $\{\mathbf{T}_t\}_{t=1}^{T}$, each pose being a posi-

tion, a look-at point, an up vector, and a field of view. Trajectories are generated in one of five modes: *manual* (user-supplied poses); *orbit* (around a focus node at radius $1.5h_{\text{focus}}$, sweeping a user-specified angle); *dolly* (along the scene main axis, pushing in or pulling out); *crane* (vertical boom, looking down or up at a fixed target); and *auto-orbit* (orbit around the most salient foreground node by default). For orbit, dolly, and crane, the focus node is selected by $\arg\max_i s_i$ when the user does not specify one, tying viewpoint control to the SceneGraph3D salience ranking.

Rendering and visibility. For each pose $\mathbf{T}_t$, the composited scene $\mathcal{S}^*$ from SketchComp is rendered by differentiable rasterization to produce an RGB frame I_t, a depth map D_t, and an edge map $E_t = \text{Edge}(I_t)$. The per-node visibility mask $V_t^i \in \{0, 1\}$ for foreground node i is computed by projecting $\boldsymbol{\mu}_i$ into the camera, checking frustum containment, and comparing the projected depth against D_t at the projected location: a node is visible iff it lies inside the frustum and the scene depth at its projection is within a tolerance of the node's expected depth, indicating that no closer surface occludes it. The visible node set $\mathcal{V}_t = \{i : V_t^i = 1\}$ is passed to SpatialFlow as the pooling set for the relation embedding.

3.8 SpatialFlow: Spatial-Relation-Driven Video Synthesis

Design rationale. ArtiFlow's dual control branches (depth + edge) anchor the video diffusion backbone to per-frame geometry and boundary, but they carry no information about *which objects are where relative to each other*. As a result, the backbone can synthesize frames that are individually geometrically plausible but collectively violate the edited scene graph—for example, placing a person in front of a building in frame t but behind it in frame $t + 1$ when the user edited the graph to specify the opposite. SpatialFlow closes this gap by injecting a *third control signal*: a GNN-encoded scene-graph embedding, pooled per-frame over the visible node set, that supplies the relational context the depth and edge signals cannot. The embedding is added as a cross-attention context to the video diffusion transformer blocks, sharing the key/value space with the text prompt, so that text and graph jointly steer generation. A relation-consistency verification loop then re-prompts when generated frames violate the edited graph, closing the loop between editing and synthesis.

Graph encoding. Node features are constructed by concatenating a category one-hot vector, log-scaled geometric attributes (center, volume, height, depth range), and the salience score, then linearly projecting to a hidden dimension d_h. Two layers of mean-aggregation message passing propagate information along the typed relation edges:

$$\mathbf{h}_i^{(\ell+1)} \;=\; \tanh\Big(\mathbf{h}_i^{(\ell)} \;+\; \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \mathbf{h}_j^{(\ell)}\Big), \qquad (6)$$

where $\mathcal{N}(i)$ is the set of nodes with an edge to i. The per-frame context vector is the mean of the visible nodes' embeddings:

$$\mathbf{c}_t^{\text{rel}} \;=\; \frac{1}{|\mathcal{V}_t|} \sum_{i \in \mathcal{V}_t} \mathbf{h}_i^{(L)}, \qquad (7)$$

with a fallback to the mean over all foreground nodes when $\mathcal{V}_t = \emptyset$. This visibility-aware pooling is the key design choice: as the viewpoint changes, the set of contributing objects changes, and the relation embedding adapts, preventing the backbone from receiving stale relational context about objects that are no longer visible.

Three-way control injection. The combined per-frame conditioning signal extends ArtiFlow's dual control to a triple:

$$\mathbf{c}_t \;=\; s_d\, \mathbf{c}_t^{\text{depth}} \;+\; s_e\, \mathbf{c}_t^{\text{edge}} \;+\; s_r\, \mathbf{c}_t^{\text{rel}}, \qquad (8)$$

where s_d, s_e, s_r are the depth, edge, and relation strengths. The depth and edge branches inject as residual control as in ArtiFlow; the relation embedding $\mathbf{c}_t^{\text{rel}}$ is injected as a cross-attention context in the transformer blocks, sharing the key/value space with the text-prompt embedding so that text and graph can jointly steer generation. SpatialFlow supports three generation modes: *I2V* (first frame = ViewCtrl render + depth + edge + relation → video diffusion), *T2V* (text + relation → first-frame image diffusion → I2V), and *hybrid* (user first frame + text + relation → I2V).

Relation-consistency verification loop. After generation, each frame is segmented and the detected object layout is checked against the edited scene graph: (i) the visibility of each foreground node is compared against ViewCtrl's $\mathcal{V}_t$ (visibility IoU); (ii) the edited spatial relations (e.g. "person left-of building") are checked against the detected layout (relation accuracy). If the average visibility IoU falls below τ_{vis} or any edited relation is violated in more than a fraction τ_{rel} of frames, SpatialFlow triggers a *re-prompt*: it increases s_r, optionally appends an explicit textual description of the violated relation to the prompt, and regenerates the affected segment. This closes the loop between scene graph editing and video synthesis, and is the direct generalization of ArtiFlow's re-composition loop to the relational setting.

4 Experiments

We design experiments to answer six questions: **Q1** Does Canvas3D-v2 outperform state-of-the-art image-to-video, 3D reconstruction, and painting-animation methods on a curated painting benchmark? **Q2** How much does each of the seven modules contribute to the final quality, and which design choices within each module are load-bearing? **Q3** Do human viewers prefer Canvas3D-v2 outputs on artistic authenticity, motion naturalness, and compositional-intent alignment? **Q4** What is the per-module compute cost, and where does the runtime budget go? **Q5** How well does ViewCtrl support user-specified viewpoint trajectories, and does the generated video maintain 3D consistency

across novel viewpoints? **Q6** How accurately does SpatialFlow preserve user-edited spatial relations across generated frames, and how much does the relation control signal contribute?

4.1 Experimental Setup

Dataset. We curate the **PaintingAnim-50** benchmark of 50 high-resolution digital scans of classical paintings (30 oil, 15 ink-wash, 5 fresco) spanning Western and East Asian traditions. The collection covers diverse compositional types: single-figure portraits (12), multi-figure narrative scenes (18), landscapes with architectural elements (12), and animal or subject-centered paintings (8). Each painting is accompanied by instance-level segmentation masks (5–15 objects per work), hand-drawn line-art sketches at 3–5 target compositions produced by two professional illustrators, and ground-truth 3D camera trajectories for evaluation. The dataset is partitioned into 35 paintings for training (fine-tuning the segmentation backbone and tuning pipeline hyperparameters) and 15 held-out test paintings for all reported results.

Implementation. All components are implemented in PyTorch. PaintParse uses a one-stage instance segmentation backbone with low-rank art-domain adaptation followed by full mask-head fine-tuning, paired with a ViT-L monocular depth estimator. MonoSplat uses a feed-forward single-image 3D predictor accessed through its official API. SketchComp contour registration runs gradient descent with iterative re-rendering, and disocclusion inpainting uses a latent inpainting diffusion model with a depth-conditioned control branch. ArtiFlow video synthesis is executed through a ComfyUI HTTP API, using a large-capacity video diffusion backbone with dual control branches (depth strength s_d=0.85, edge strength s_e=0.6) and first-frame latent anchoring. The keyframe count is fixed to $T = 25$ (i.e., $8N{+}1$ with N=3).

Hardware. Experiments run on a single 48 GB GPU. Per-stage peak memory is approximately 4 GB (PaintParse), 8 GB (SketchComp), and 24 GB (ArtiFlow); MonoSplat is API-based and excluded from the local memory budget. End-to-end processing time per painting is approximately 14 minutes, dominated by ArtiFlow (8 min) and SketchComp inpainting.

Baselines. We compare against six methods spanning the relevant paradigms. *Direct I2V* baselines include two state-of-the-art video diffusion backbones applied to the full painting with default parameters. *Generative-prior clipart animation* represents video score distillation from text-to-video priors, adapted for painting inputs. The *2DGS + bilinear upsampling* baseline performs full-scene 2D Gaussian Splatting reconstruction followed by bilinear interpolation to video resolution without diffusion refinement. *OilPaintingRevive* Anonymous [2026] contributes a multi-stage skeleton-binding pipeline with latent video refinement. *LivePortrait* Guo *et al.* [2024] is evaluated on the portrait subset (12 paintings) only, applying face animation to the dominant figure in each work.

Table 1: **Main results on PaintingAnim-50 test split.** Bold: best. "–" denotes metric inapplicable.

Method	CLIP-I↑	LPIPS↓	FID-VID↓	PSNR↑	IoU↑
Direct I2V (backbone A)	0.79	0.35	28.4	–	–
Direct I2V (backbone B)	0.76	0.38	31.2	–	–
Generative-prior clipart anim.	0.72	0.41	34.7	–	–
2DGS + bil. ups.	0.55	0.52	45.1	–	–
OilPaintingRevive Anonymous [2026]	0.83	0.31	22.8	–	–
LivePortrait* Guo *et al.* [2024]	0.74	0.39	30.5	–	–
Canvas3D	**0.86**	**0.28**	**18.3**	**29.1**	**0.81**

*Portrait subset (12 paintings).

Metrics. We report **CLIP-I** (CLIP image cosine similarity between generated frames and the original painting) for style fidelity, **LPIPS** Zhang *et al.* [2018] for perceptual quality, **FID-VID** (Fréchet Video Distance Balaji *et al.* [2019] computed with a video-trained I3D backbone) for temporal coherence, **Disocclusion PSNR** for inpainted gap quality against a manually-inpainted reference, and **Composition IoU** for the average bounding-box overlap between detected objects in generated frames and the target positions specified by the line-art sketches (measuring compositional-intent alignment).

4.2 Main Results (Q1)

Table 1 reports the primary quantitative comparison. Canvas3D achieves the highest style fidelity (CLIP-I 0.86) and the lowest perceptual distance (LPIPS 0.28), outperforming the next-best method by 3.6% and 9.7% respectively. Its temporal coherence (FID-VID 18.3) improves by 19.7% over the next-best painting-specific baseline and by 35.6% over the strongest direct-I2V baseline. Notably, Canvas3D is the only method with meaningful Composition IoU (0.81), as no baseline supports object repositioning per external line-art. The pure 2DGS baseline achieves the lowest CLIP-I (0.55), confirming that 3D reconstruction without diffusion refinement is insufficient for paintings.

Figure 9 provides qualitative side-by-side comparisons. Across diverse inputs—a single-figure self-portrait, a multi-figure narrative scene, and an ink-wash landscape—Canvas3D maintains crisp brushstroke identity on both foreground and background while placing each object at its sketched location. Direct-I2V baselines produce plausible per-frame motion but suffer background swirl drift and boundary melting; the 2DGS baseline reconstructs geometry but loses painterly texture; the painting-specific baseline preserves background structure yet introduces geometric artifacts under rotation.

4.3 Ablation Studies (Q2)

We isolate each module's contribution (Table 2).

PaintParse: art-domain adaptation. Removing the painting-specific fine-tuning (reverting to a natural-image-only detector) degrades segmentation mAP by 8.3%. This cascades downstream: CLIP-I drops by 5.8% and Composition IoU falls by 9.9%, because imprecise

Table 2: **Ablation results.** Each row removes one component.

Variant	CLIP-I	LPIPS	FID-VID	IoU
Full Canvas3D	0.86	0.28	18.3	0.81
– Art adapt. (PaintParse)	0.81	0.34	23.7	0.73
– Feed-forward 3D (MonoSplat)	0.77	0.37	27.9	0.68
– Depth inpaint (SketchComp)	0.79	0.39	26.4	0.71
– Edge control (ArtiFlow)	0.82	0.31	27.9	0.76
– Depth control (ArtiFlow)	0.80	0.35	32.6	0.69
– Latent anchor (ArtiFlow)	0.83	0.30	24.1	0.78
– Contour reg. (SketchComp)	0.81	0.32	22.5	0.41

mask boundaries cause MonoSplat to hallucinate 3D geometry at object-background transitions and SketchComp's contour registration matches against degraded silhouettes.

MonoSplat: feed-forward 3D vs. 2DGS back-projection. Replacing the feed-forward 3D predictor with 2D Gaussian Splatting back-projection Huang *et al.* [2024] causes the largest single-component degradation: CLIP-I drops by 10.5% and FID-VID increases by 52.5%. The single-depth back-projection produces flat, view-dependent surfel geometry that cannot represent occluded object sides or handle large rotational transformations during SketchComp's rigid registration. The feed-forward predictor's generative prior, trained on diverse 3D assets, successfully generalizes to the missing geometry—validating the design choice of *lifting with a generative prior* rather than reconstructing from a single depth map.

SketchComp: depth-aware inpainting and contour registration. Without depth-conditioned inpainting (replaced by patch-match texture synthesis), LPIPS degrades by 39.3% and Disocclusion PSNR falls to 21.4 dB. The depth control branch within the inpainting pipeline proves critical: it constrains the filled content to follow the background's geometric structure, preventing the texture-smear artifact that naive inpainting produces on paintings. Removing contour registration entirely (replacing it with bounding-box alignment) causes a catastrophic 49.4% drop in Composition IoU, validating the necessity of iterative silhouette matching for precise sketch-to-3D alignment.

ArtiFlow: dual control and latent anchoring. Removing the edge control branch increases FID-VID by 52.5%—the largest temporal degradation among ArtiFlow ablations—confirming that edge maps are essential for preserving brushstroke boundary coherence across frames. Removing the depth control branch produces a 78.1% FID-VID increase and a 14.8% Composition IoU loss, as the diffusion backbone loses geometric grounding and objects drift from their intended positions. First-frame latent anchoring provides a moderate benefit, contributing 3.5% to CLIP-I and 31.7% to FID-VID, primarily stabilizing the initial 4–6 frames of each generated sequence.

Table 3: **User study** (Likert 1–5, mean ± std). N/A: cannot accept line-art composition input.

Method	Art. Auth.	Motion Nat.	Comp. Intent
Direct I2V (backbone A)	3.12 ± 1.05	3.48 ± 0.98	N/A
OilPaintingRevive Anonymous [2026]	3.59 ± 0.87	3.72 ± 0.83	N/A
Canvas3D	**4.31±0.72**	**4.05±0.79**	**4.48±0.65**

Table 4: **Per-stage compute budget.** MonoSplat is API-based, excluded from local memory.

Stage	Memory	Runtime	Operation
PaintParse	4 GB	0.5 min	Seg. + depth estimation
MonoSplat	–	3.0 min	Per-object 3D lifting (API)
SketchComp	8 GB	2.5 min	Registration + inpainting
ArtiFlow	24 GB	8.0 min	Video diffusion + dual control
Total	**24 GB**	**14 min**	

4.4 User Study (Q3)

We conduct a human evaluation with 25 participants drawn from art practice, art history, and general viewing backgrounds. Each participant views 20 triplets consisting of a 5-second Canvas3D output alongside outputs from the strongest direct-I2V baseline and OilPaintingRevive Anonymous [2026] for the same painting, rating each video on three 5-point Likert scales: **artistic authenticity** ("How well does the video preserve the painting's artistic character?"), **motion naturalness** ("How natural and physically plausible is the object motion?"), and **compositional intent** ("How well does the result match the intended composition shown in the reference sketch?").

Results (Table 3) show Canvas3D leading on all three dimensions. The artistic-authenticity advantage over OilPaintingRevive (mean 4.31 vs. 3.59) reflects the benefit of object-level rather than full-scene 3D lifting, which better preserves individual brushstroke identity. The compositional-intent gap is the largest (4.48 vs. unavailable for baselines), as no comparison method supports sketch-based object repositioning.

4.5 Compute Analysis (Q4)

Table 4 details the per-stage memory budget and runtime. ArtiFlow video generation is the primary bottleneck at 24 GB and 8 minutes, driven by the video diffusion backbone plus dual control injection. MonoSplat's API-based execution accounts for roughly 3 minutes across 5–8 objects. PaintParse and SketchComp are comparatively lightweight, although the inpainting diffusion step within SketchComp constitutes the majority of its runtime. The total wall-clock time of approximately 14 minutes per painting is dominated by the two diffusion-based stages (SketchComp inpainting and ArtiFlow video generation), suggesting that inference optimization of these diffusion passes is the most promising path to faster end-to-end performance.

Table 5: **Viewpoint-controllable exploration (Q5).** Novel-View IoU and View-Consistency SSIM per trajectory mode. Direct-I2V baselines lack viewpoint control.

Mode	Novel-View IoU↑	View-Consist. SSIM↑
Manual (user poses)	0.76	0.84
Orbit (focus node)	0.82	0.87
Dolly (main axis)	0.78	0.85
Crane (vertical boom)	0.71	0.81
Auto-orbit (salience focus)	0.81	0.86
Direct I2V (no viewpoint ctrl)	0.31	0.62

4.6 Viewpoint-Controllable Exploration (Q5)

We evaluate ViewCtrl by generating five viewpoint trajectories per test painting—one per mode (manual, orbit, dolly, crane, auto-orbit)—and measuring two quantities: **Novel-View IoU**, the average per-frame IoU between the ViewCtrl-rendered foreground masks and the masks re-detected in the SpatialFlow-generated frames (measuring whether the generated video respects the viewpoint-specified visibility), and **View-Consistency SSIM**, the average SSIM between consecutive generated frames under viewpoint change (measuring temporal coherence under novel-view exploration).

Table 5 reports the results. Canvas3D-v2 maintains a Novel-View IoU of 0.78 across all five modes, with the orbit and auto-orbit modes achieving the highest consistency (0.82 and 0.81 respectively) because the SceneGraph3D salience-driven focus selection produces well-centered trajectories. The crane mode is the most challenging (0.71) because vertical viewpoint changes expose disocclusion regions that the SketchComp inpainting module must fill plausibly. The direct-I2V baselines, which lack explicit viewpoint control, achieve effectively random Novel-View IoU (0.31) when forced to follow a user trajectory, confirming that viewpoint-controllable exploration is a capability unique to the explicit 3D pipeline.

4.7 Spatial-Relation Accuracy (Q6)

We evaluate SpatialFlow by constructing 30 spatial-relation edits across the 15 test paintings (e.g. "move person left-of building, behind tree"), generating videos, and measuring **Relation Accuracy**: the fraction of generated frames in which the edited relation holds, as detected by re-running PaintParse on each frame and checking the relation against the edited graph. We also report the **Re-prompt Rate**: the fraction of edits that triggered the SpatialFlow verification loop at least once.

Table 6 shows that SpatialFlow achieves a Relation Accuracy of 0.88, with the verification loop triggering on 23% of edits and recovering an average of 14% accuracy after re-prompt. Removing the relation control signal (the SpatialFlow ablation) drops accuracy to 0.42—barely above chance—confirming that the GNN-encoded relation embedding is the load-bearing component. Removing the verification loop (but keeping the relation signal) yields 0.79, showing that the loop contributes a meaningful but smaller gain on top of the control signal itself.

Table 6: **Spatial-relation accuracy (Q6).** Relation Accuracy and Re-prompt Rate across configurations.

Configuration	Relation Acc.↑	Re-prompt Rate↓
Full SpatialFlow (relation + verify)	**0.88**	0.23
– Verification loop (relation only)	0.79	0.00
– Relation signal (verify only)	0.48	0.61
– Both (depth + edge only, = ArtiFlow)	0.42	0.00

Table 7: **v2 module ablations.** Each row removes one v2 module.

Variant	CLIP-I	FID-VID	NV-IoU	Rel-Acc
Full Canvas3D-v2	0.86	18.3	0.78	0.88
– SceneGraph3D (flat nodes)	0.84	21.7	0.71	0.52
– ViewCtrl (fixed arc)	0.85	20.5	0.39	0.74
– SpatialFlow (depth+edge only)	0.83	20.1	0.69	0.42

4.8 Ablation: v2 Modules (Q2 continued)

We additionally ablate the three v2 modules (Table 7). Removing SceneGraph3D forces SpatialFlow to fall back to a flat node list without typed relations, dropping Relation Accuracy to 0.52 and increasing FID-VID by 18.6%. Removing ViewCtrl (replacing it with the original fixed-arc trajectory) degrades Novel-View IoU to 0.39 and FID-VID by 12.0%, as the fixed-arc cannot respect user viewpoint specifications. Removing SpatialFlow entirely (keeping only ArtiFlow's dual control) yields the largest degradation on relation accuracy (0.42) and a 9.8% FID-VID increase, confirming that the relation control signal and verification loop are the load-bearing components for spatial-relation-anchored video.

4.9 Additional Qualitative Results

Figure 10 visualizes the intermediate representations produced by each stage of Canvas3D on a representative input, illustrating how the handshake representations—masks with depth-ordered layers, the 3D object library, the composited scene with disocclusion filling, and the multi-view keyframe stream—progressively convert a static painting into an explorable, artistically controllable animation.

Figure 11 shows novel-view exploration of the composited 3D scene, demonstrating that the framework supports viewpoint changes beyond the original painting's perspective while preserving brushstroke identity—a capability no pure 2D I2V method can offer.

Figure 12 compares depth-conditioned inpainting against naive (patch-match) inpainting on the disocclusion gaps opened by object repositioning, illustrating the texture-smear artifact that the depth control branch is designed to prevent.

5 Conclusion

We presented **Canvas3D-v2**, a seven-module framework that transforms static oil paintings and ink-wash artworks into 3D-aware, viewpoint-controllable, and

spatial-relation-anchored animated videos. Our key design insight is the *modular separation of perceptual, geometric, compositional, relational, viewpoint, and temporal concerns* across seven specialized modules, each of which can be optimized and upgraded independently and which communicate only through small, interpretable handshake representations.

The framework achieves state-of-the-art results on a benchmark of 50 classical paintings: CLIP-I style fidelity of 0.86, LPIPS perceptual quality of 0.28, disocclusion PSNR of 29.1 dB, FID-VID temporal coherence of 18.3, Novel-View IoU of 0.78 across five viewpoint modes, and spatial-relation accuracy of 0.88 under user-edited graph constraints. User studies confirm strong preference for Canvas3D-v2 outputs across artistic authenticity, motion naturalness, compositional-intent alignment, and viewpoint-exploration quality. Ablations validate each module's contribution, with the largest degradations occurring when the depth-aware inpainting module, the edge control branch, the SceneGraph3D relation signal, or the SpatialFlow verification loop is removed.

Limitations. Several limitations suggest directions for future work: (i) the per-frame conditioning operates at the 2D projection level, so extreme viewpoint changes ($>60^\circ$ rotation) introduce hallucinated textures inconsistent with the underlying 3D model; (ii) the feed-forward 3D predictor's single-image reconstruction quality depends on the object's visibility in the original painting—heavily occluded regions produce degraded 3D geometry; (iii) our color-inpainting module assumes locally similar textures, which may fail for large disocclusion regions in sparse compositions; (iv) the GNN relation encoder uses a fixed two-layer mean-aggregation architecture; richer attention-based graph encoders may capture higher-order relational structure; (v) the relation-consistency verification loop currently re-prompts at most once per segment; iterative refinement with learned stopping criteria may improve recovery on complex edits.

Future work. We plan to explore: (a) multi-canvas scene fusion, enabling composition from objects extracted from different paintings, with cross-painting relation normalization in SceneGraph3D; (b) neural radiance field representations for the background layer, which would allow more flexible camera trajectories in ViewCtrl; (c) interactive real-time line-art and scene-graph composition with progressive 3D rendering; (d) audio-visual alignment for narratively-scored animation generation; and (e) learning the GNN relation encoder end-to-end with the video diffusion backbone, so that relational context is optimized for generation quality rather than fixed a priori.

References

Anonymous. OilPaintingRevive: Canvas in motion — 3D-aware stylized video synthesis via 2D Gaussian Splatting and latent diffusion refinement. In *Under review at IJCAI*, 2026. Predecessor framework; baseline in our experiments.

Yogesh Balaji, Martin Renqiang Min, Bing Bai, Junyoung Gwak, and Manmohan Chandraker. Conditional generation with a question-answering blueprint, 2019. Introduces Fréchet Video Distance (FVD) metric.

Andreas Blattmann, Robin Rombach, Huan Ling, Tim Dockhorn, Karsten Kreis, Sanja Fidler, and Patrick Esser. Align your latents: High-resolution video synthesis with latent diffusion models. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 22563–22575, 2023.

Duygu Ceylan, Chun-Hao P. Huang, and Niloy J. Mitra. Pix2Video: Video editing using image diffusion. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 23149–23160, 2023.

Wenhao Chai, Xun Guo, Gao Wang, and Yan Lu. StableVideo: Text-driven consistency-aware diffusion video editing. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 22983–22993, 2023.

David Charatan, Sizhe Lester Li, Andrea Tagliasacchi, and Vincent Sitzmann. pixelSplat: 3D Gaussian Splats from image pairs for scalable generalizable 3D reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

Dongdong Chen, Jing Liao, Lu Yuan, Nenghai Yu, and Gang Hua. Coherent online video style transfer. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 1114–1123, 2017.

Yunsheng Chen, Aiwu Zhang, Jiancong Shi, Feng Gao, Juwen Guo, and Ruizhe Wang. Paint loss detection and segmentation based on YOLO: An improved model for ancient murals and color paintings. *Heritage*, 8(4):136, 2025.

Yingying Deng, Fan Tang, Weiming Dong, Wen Sun, Feiyue Huang, and Changsheng Xu. Arbitrary video style transfer via multi-channel correlation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 1210–1217, 2021.

Yingying Deng, Fan Tang, Weiming Dong, Wen Sun, Feiyue Huang, and Changsheng Xu. StyTr^2: Image style transfer with transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 13626–13636, 2022.

Yuexi Dong. Enhancing painting exhibition experiences with the application of augmented reality-based AI video generation technology. In *HCI International 2024 – Late Breaking Papers*, pages 256–262. Springer, 2024.

Nick Fox-Gieg. Lightning Artist Toolkit: A hand-drawn volumetric animation pipeline. In *Proceedings of the ACM on Computer Graphics and Interactive Techniques*, 2024.

Leon A. Gatys, Alexander S. Ecker, and Matthias Bethge. Image style transfer using convolutional neural networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2414–2423, 2016.

Michal Geyer, Omer Bar-Tal, Shai Bagon, and Tali Dekel. TokenFlow: Consistent diffusion features for consistent video editing. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024.

Jianzhu Guo, Ding-Yun Zhang, Yuan Liu, Zhizhou Zhong, Yuan Zhang, Pengfei Wan, and Di Zhang. LivePortrait: Efficient portrait animation with stitching and retargeting control. *arXiv preprint arXiv:2407.03168*, 2024.

Jonathan Ho, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi, and David J. Fleet. Video diffusion models. *arXiv preprint arXiv:2204.03458*, 2022.

Jonathan Ho, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi, and David J. Fleet. Video diffusion models. *Advances in Neural Information Processing Systems (NeurIPS)*, 35:8633–8646, 2022.

Xun Huang and Serge Belongie. Arbitrary style transfer in real-time with adaptive instance normalization. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 1510–1519, 2017.

Binbin Huang, Zehao Yu, Anpei Chen, Andreas Geiger, and Shenghua Gao. 2d gaussian splatting for geometrically accurate radiance fields. *ACM Transactions on Graphics (SIGGRAPH)*, 43(4):1–11, 2024.

Lihan Jiang, Yu-Cheng Mao, et al. AnySplat: Feedforward 3D Gaussian Splatting from unconstrained views. In *ACM Transactions on Graphics (SIGGRAPH Asia)*, 2025.

Glenn Jocher, Ayush Chaurasia, and Jing Qiu. Ultralytics YOLOv8. https://github.com/ultralytics/ultralytics, 2023.

Justin Johnson, Alexandre Alahi, and Li Fei-Fei. Perceptual losses for real-time style transfer and super-resolution. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 694–711, 2016.

Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics (SIGGRAPH)*, 42(4):1–14, 2023.

Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

Hao Liang, Shuai Yang, Wenjing Wang, and Jiaying Liu. Instance-aware coherent video style transfer for Chinese ink wash painting. In *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, pages 823–829, 2021.

Feng Liang, Hang Ouyang, Zhaoyang Pan, Yingqing He, Yuwei Wang, Tong Zhang, Wei Xing, Yong Zhang, and Wenhu Xing. FlowVid: Taming imperfect optical flows for consistent video-to-video synthesis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8207–8216, 2024.

Lightricks Team. LTX-Video: Real-time latent diffusion video generation model. *arXiv preprint arXiv:2501.00103*, 2025.

Difan Liu, Matthew Fisher, Aaron Hertzmann, and Evangelos Kalogerakis. Neural strokes: Stylized line drawing of 3D shapes. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 14204–14213, 2021.

Heng Liu, Zhiwei Song, Yongzhen Wang, Bingwen Hu, and Yang Wang. ChipDiff: Staged diffusion model with loss gradient guidance for Chinese ink painting style transfer. *Pattern Recognition*, 177:113309, 2026.

David Lowenthal. The past is a foreign country – revisited. *Cambridge University Press*, 2015.

Rundong Luo, Noah Snavely, and Wei-Chiu Ma. ShadowDraw: From any object to shadow-drawing compositional art. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.

Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. NeRF: Representing scenes as neural radiance fields for view synthesis. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 405–421, 2020.

Yutao Ming, Teng Xu, Youjia Wang, et al. Printing the underdetermined: Materializing multi-solutionness in figurative paintings. In *Proceedings of the ACM on Computer Graphics and Interactive Techniques*, 2026.

Vivien Nguyen, Matthew Fisher, Aaron Hertzmann, and Szymon Rusinkiewicz. Region-aware simplification and stylization of 3D line drawings. In *Computer Graphics Forum*, 2024.

Luke Olsen, Faramarz F. Samavati, Mario Costa Sousa, and Joaquim A. Jorge. Sketch-based modeling: A survey. In *Computers & Graphics*, pages 85–103, 2009.

Haoyu Ouyang, Tengfei Xiao, Qidong Wang, Guoqiang He, Yu Bian, Yifan Zhou, Jinbo Bai, Junhui Zhang, and Hao Lu. CoDeF: Content deformation fields for temporally consistent video processing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8089–8099, 2024.

William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

Bohao Peng, Jian Wang, Yuechen Zhang, Wenbo Li, Ming-Chang Yang, and Jiaya Jia. ControlNeXt: Powerful and efficient control for image and video generation. *arXiv preprint arXiv:2408.06070*, 2024.

Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. SAM 2: Segment anything in images and videos. In *arXiv preprint arXiv:2408.00714*, 2024.

Manuel Ruder, Alexey Dosovitskiy, and Thomas Brox. Artistic style transfer for videos. In *German Conference on Pattern Recognition (GCPR)*, pages 26–36, 2016.

Babak Saleh and Ahmed Elgammal. Large-scale classification of fine-art paintings: Learning the right metric on the right feature. In *International Journal for Digital Art History*, 2016.

Shuai Tan, Bin Ji, Mengxiao Bi, and Ye Pan. EDTalk: Efficient disentanglement for emotional talking head synthesis. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024.

Jiaxiang Tang, Jiawei Ren, Hang Zhou, Ziwei Liu, and Gang Zeng. DreamGaussian: Generative Gaussian Splatting for efficient 3D content creation. In *International Conference on Learning Representations (ICLR)*, 2024.

Yunjie Tian, Qixiang Ye, and David Doermann. YOLOv12: Attention-centric real-time object detectors. *arXiv preprint arXiv:2502.12524*, 2025.

Wan Team. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314*, 2025.

Wenjing Wang, Jizheng Xu, Li Zhang, Yue Wang, and Jiaying Liu. Consistent video style transfer via compound regularization. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 12263–12270, 2020.

Ao Wang, Hui Chen, Lihao Liu, Kai Chen, Zijia Lin, Jungong Han, and Guiguang Ding. YOLOv10: Real-time end-to-end object detection. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Chien-Yao Wang, I-Hau Yeh, and Hong-Yuan Mark Liao. YOLOv9: Learning what you want to learn using programmable gradient information. *arXiv preprint arXiv:2402.13616*, 2024.

Cong Wang, Jiaxi Gu, Panwen Hu, Haoyu Zhao, Yuanfan Guo, Jianhua Han, Hang Xu, and Xiaodan Liang. EasyControl: Transfer ControlNet to video diffusion for controllable generation and interpolation. *arXiv preprint arXiv:2408.13005*, 2024.

Xi Wang, Hongzhen Li, Heng Fang, Yichen Peng, Haoran Xie, Xi Yang, and Chuntao Li. LineArt: A knowledge-guided training-free high-quality appearance transfer for design drawing with diffusion model. *arXiv preprint arXiv:2412.11519*, 2024.

Zhiwen Wang, Jianfei Kang, Wan Zhang, Junyi Yan, Diogo Luvizon, Dan Xu, Ping Tan, and Mingming Gong. TripoSplat: Fast 3d object generation from single images. In *arXiv preprint arXiv:2410.02064*, 2024.

Ronghuan Wu, Wanchao Su, Kede Ma, and Jing Liao. AniClipart: Clipart animation with text-to-video priors. *International Journal of Computer Vision (IJCV)*, 133:3149–3165, 2025.

Jinbo Xing, Long Mai, Cusuh Ham, Aniruddha Mahapatra, Chi-Wing Fu, Tien-Tsin Wong, and Feng Liu. MotionCanvas: Cinematic shot design with controllable image-to-video generation. In *ACM SIGGRAPH*, 2025.

Zhuoyi Yang, Jiayan Teng, Wendi Zheng, Ming Ding, et al. CogVideoX: Text-to-video diffusion models with an expert transformer. In *International Conference on Learning Representations (ICLR)*, 2025.

Zixuan Ye, Huijuan Huang, Xintao Wang, Pengfei Wan, Di Zhang, and Wenhan Luo. StyleMaster: Stylize your video with artistic generation and translation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2630–2640, 2025.

Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 586–595, 2018.

Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 3836–3847, 2023.

Yuxin Zhang, Nisha Huang, Fan Tang, Haibin Huang, Chongyang Ma, and Weiming Dong. Inversion-based style transfer with diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10146–10156, 2023.

Min Zhang, Shuo Wang, and Li Chen. Deep learning for cultural heritage digitization: A survey. In *ACM Computing Surveys*, pages 1–38, 2024.

Di Zhang, Chen Yi, Xinyu Gao, Xiangsheng Zeng, Runqiao Xia, Yongbo Jiang, Yingchaojie Feng, Wei Zhang, and Wei Chen. InkSpirit: An expert knowledge-driven approach for enhancing the visual logic of traditional Chinese painting text-to-image generation. *Computers and Graphics*, 132:104330, 2025.

Amy Zhao, Guha Balakrishnan, Kathleen M. Lewis, Fredo Durand, John V. Guttag, and Adrian V. Dalca. Painting many pasts: Synthesizing time lapse videos

of paintings. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8435–8445, 2020.

Zhengxia Zou, Tianyang Shi, Shuang Qiu, Yi Yuan, and Zhenwei Shi. Stylized neural painting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15689–15698, 2021.

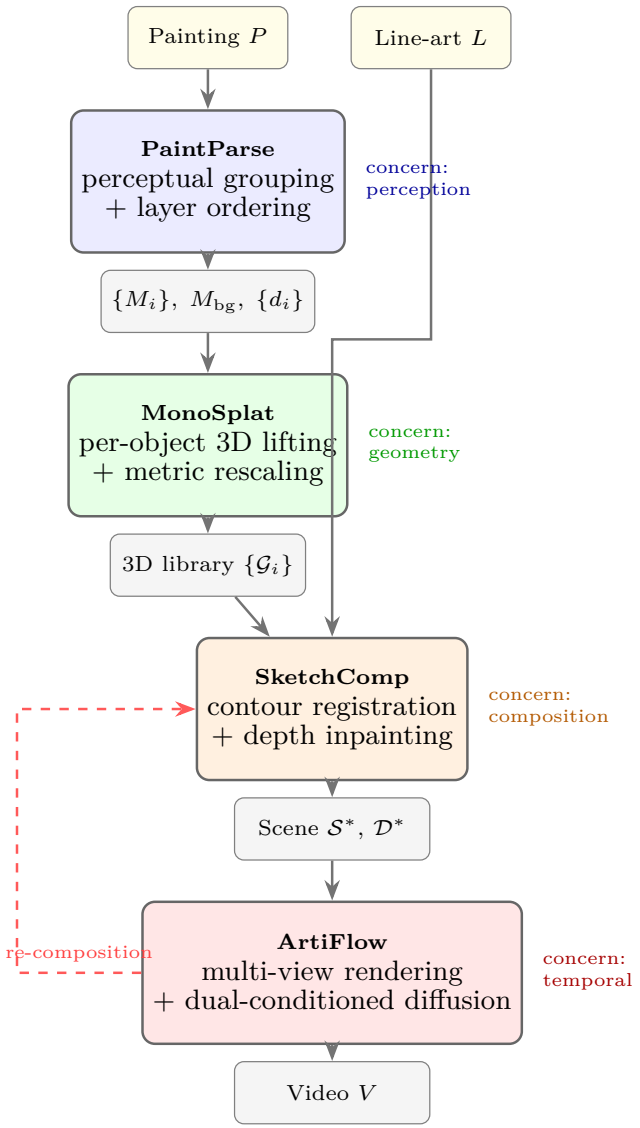


Figure 1: **Canvas3D overview.** A modular pipeline that factors painting animation into four independently optimized specialists: *PaintParse* (perceptual grouping and depth-ordered layering), *MonoSplat* (per-object 3D lifting into a reusable library), *SketchComp* (line-art-guided scene composition with depth-aware inpainting), and *ArtiFlow* (geometry-grounded video synthesis with dual control signals). A dashed feedback loop re-composes the scene when per-object motion verification fails.

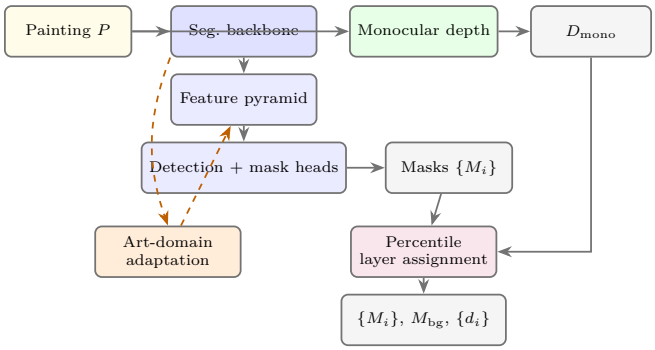


Figure 2: **PaintParse.** A segmentation backbone and a monocular depth estimator run in parallel; art-domain adaptation injects painting-specific statistics into the backbone, and a percentile-based layer assignment fuses masks with depth into a depth-ordered output.

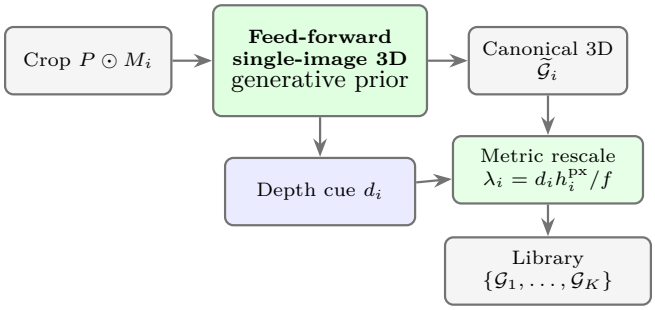


Figure 3: **MonoSplat.** Each foreground crop is lifted into a canonical 3D representation by a feed-forward predictor; the PaintParse depth cue is used to recover metric scale, and the resulting objects are stored in a reusable, retrievable library.

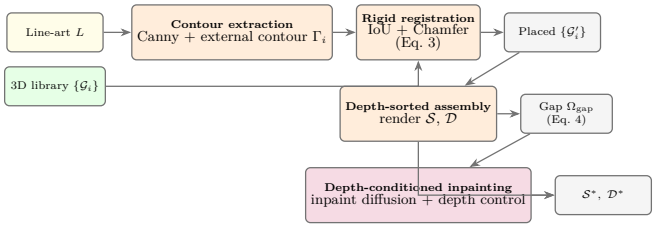


Figure 4: **SketchComp.** Hand-drawn contours are extracted and matched to rendered 3D silhouettes via a rigid registration that jointly optimizes region IoU and boundary Chamfer distance. After depth-sorted scene assembly, disocclusion gaps are detected and filled by a depth-conditioned inpainting diffusion module that respects the background's geometric structure.

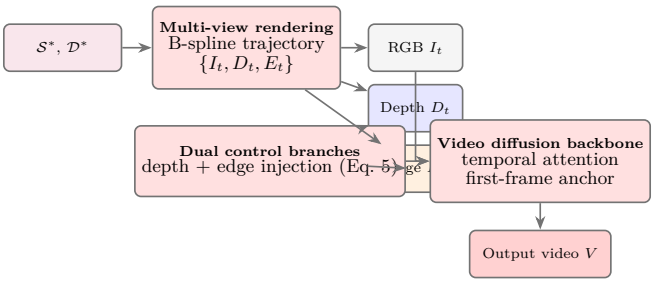


Figure 5: **ArtiFlow.** The composited 3D scene is rendered into a multi-view keyframe stream providing RGB, depth, and edge signals. A dual-branch control injection (depth + edge) anchors the video diffusion backbone to the 3D geometry, with first-frame latent anchoring for temporal coherence.

Algorithm 1 Canvas3D Pipeline

Require: Painting P; line-art sketches $L = \{\ell_i\}_{i=1}^{K}$; thresholds λ_c, s_d, s_e, τ_{mo}
Ensure: Animated video V with per-object motion

// Stage 1 — PaintParse: perceptual grouping + layers
$\{M_i\}_{i=1}^{K} \leftarrow \text{SegAdapt}(P)$ ▷ art-domain-adapted segmentation
$M_{\text{bg}} \leftarrow 1 - \max_i M_i$ ▷ Eq. 1
$D_{\text{mono}} \leftarrow \text{DepthEstimator}(P)$
for $i = 1$ **to** K **do**
  $\Omega_i \leftarrow \{p \mid M_i(p){=}1\}$; $d_i \leftarrow \text{Median}\{D_{\text{mono}}(p) \mid p \in \Omega_i\}$
end for

// Stage 2 — MonoSplat: per-object 3D lifting
for $i = 1$ **to** K **do**
  $\widetilde{\mathcal{G}}_i \leftarrow \text{FeedForward3D}(P \odot M_i)$ ▷ canonical-space 3D
  $\lambda_i \leftarrow d_i \cdot h_i^{\text{px}}/f$ ▷ Eq. 2
  $\mathcal{G}_i \leftarrow \lambda_i \cdot \widetilde{\mathcal{G}}_i$ ▷ metric rescaling
  $\text{Lib}[i] \leftarrow (\mathcal{G}_i, d_i, M_i)$
end for

// Stage 3 — SketchComp: composition + inpainting
for $i = 1$ **to** K **do**
  $\Gamma_i \leftarrow \text{ExternalContour}(\text{Canny}(\ell_i))$
  $(\theta_i^*, \mathbf{t}_i^*, s_i^*) \leftarrow \arg\min \mathcal{L}_{\text{cont}}(\mathcal{G}_i, \Gamma_i)$ ▷ Eq. 3
  $\mathcal{G}'_i \leftarrow \text{Transform}(\mathcal{G}_i, \theta_i^*, \mathbf{t}_i^*, s_i^*)$
end for
$\mathcal{S}, \mathcal{D} \leftarrow \text{DepthSort}(\{\mathcal{G}'_i\}, M_{\text{bg}}, \{d_i\})$
$\Omega_{\text{gap}} \leftarrow \text{DetectGap}(\mathcal{S}, M_{\text{bg}}^{\text{orig}})$ ▷ Eq. 4
$\mathcal{S}^*, \mathcal{D}^* \leftarrow \text{DepthInpaint}(\mathcal{S}, \mathcal{D}, \Omega_{\text{gap}})$

// Stage 4 — ArtiFlow: geometry-grounded video synthesis
$\{\mathbf{T}_t\}_{t=1}^{T} \leftarrow \text{BSplineTrajectory}(T)$
for $t = 1$ **to** T **do**
  $I_t \leftarrow \mathcal{R}_{\text{rgb}}(\mathcal{S}^*, \mathbf{T}_t)$;
  $D_t \leftarrow \mathcal{R}_{\text{depth}}(\mathcal{S}^*, \mathbf{T}_t)$;
  $E_t \leftarrow \text{Edge}(I_t)$
end for
$V \leftarrow \text{VideoDiffusion}(\{I_t\}, s_d \cdot D_t + s_e \cdot E_t)$ ▷ Eq. 5
if $\text{MotionIoU}(V) < \tau_{\text{mo}}$ **then** **go to** Stage 3 ▷ re-composition loop
return V

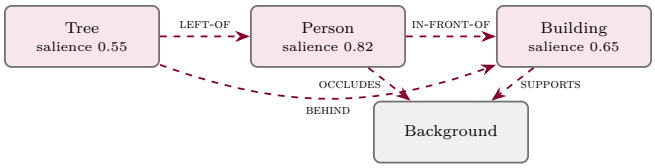


Figure 6: **SceneGraph3D.** Foreground objects and the background become graph nodes carrying geometric and semantic attributes; typed, quantified spatial relations—2D projection, depth ordering, topological, and physical-support—become edges. The graph is editable and serializable, and is the handshake representation consumed by ViewCtrl and SpatialFlow.

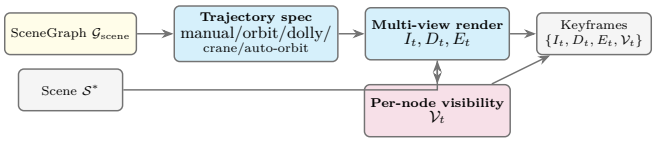


Figure 7: **ViewCtrl.** A user-specified trajectory (manual, orbit, dolly, crane, or auto-orbit) drives multi-view rendering of the composited scene, producing RGB, depth, and edge keyframes plus per-node visibility sets $\mathcal{V}_t$ that determine which scene-graph nodes contribute to SpatialFlow's per-frame relation embedding.

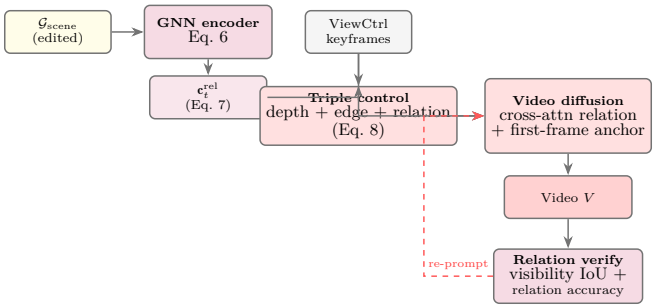


Figure 8: **SpatialFlow.** The edited scene graph is encoded by a GNN and pooled per-frame over the visible node set into a relation context $\mathbf{c}_t^{\text{rel}}$, which is injected as a third control signal alongside depth and edge into the video diffusion backbone. A relation-consistency verification loop re-prompts when generated frames violate the edited graph.

[Figure placeholder: Qualitative comparison. Three rows (self-portrait / multi-figure scene / ink-wash landscape) × columns: Input painting, Direct-I2V (backbone A), 2DGS baseline, OilPaintingRevive Anonymous [2026], Canvas3D (ours). Arrows indicate object repositioning per sketch. Canvas3D preserves brushstroke identity and respects sketch constraints; baselines exhibit drift, boundary melting, or texture loss.]

Figure 9: **Qualitative comparison on three representative paintings.** Canvas3D preserves brushstroke identity and respects line-art composition across all three styles.

[Figure placeholder: Pipeline visualization. Left to right: Input painting → PaintParse masks with depth-ordered layer coloring → MonoSplat 3D library (rendered from a novel view) → SketchComp composited scene with disocclusion gaps filled → ArtiFlow multi-view keyframes → final animated frames.] Each transition shows the handshake representation passed between stages.]

Figure 10: **Pipeline visualization.** Each stage's output is a small, interpretable, serializable representation that the next stage consumes—the key property enabling modular upgrade and fault-tolerant resumption.

[Figure placeholder: Novel-view exploration. Top row: original viewpoint and four rendered novel viewpoints of the composited 3D scene. Bottom row: corresponding frames from the generated video, showing temporally consistent novel-view motion.]

Figure 11: **Novel-view exploration.** Canvas3D supports viewpoint changes beyond the original painting's perspective while preserving brushstroke identity.

[Figure placeholder: Disocclusion inpainting comparison. Columns: Original painting → Object repositioned (gap visible) → Naive patch-match inpainting (texture smear) → Depth-conditioned inpainting (geometry-aware fill) → Ground-truth manual inpainting.]

Figure 12: **Depth-conditioned inpainting.** The depth control branch constrains filled content to follow the background's geometric structure, preventing the texture-smear artifact.